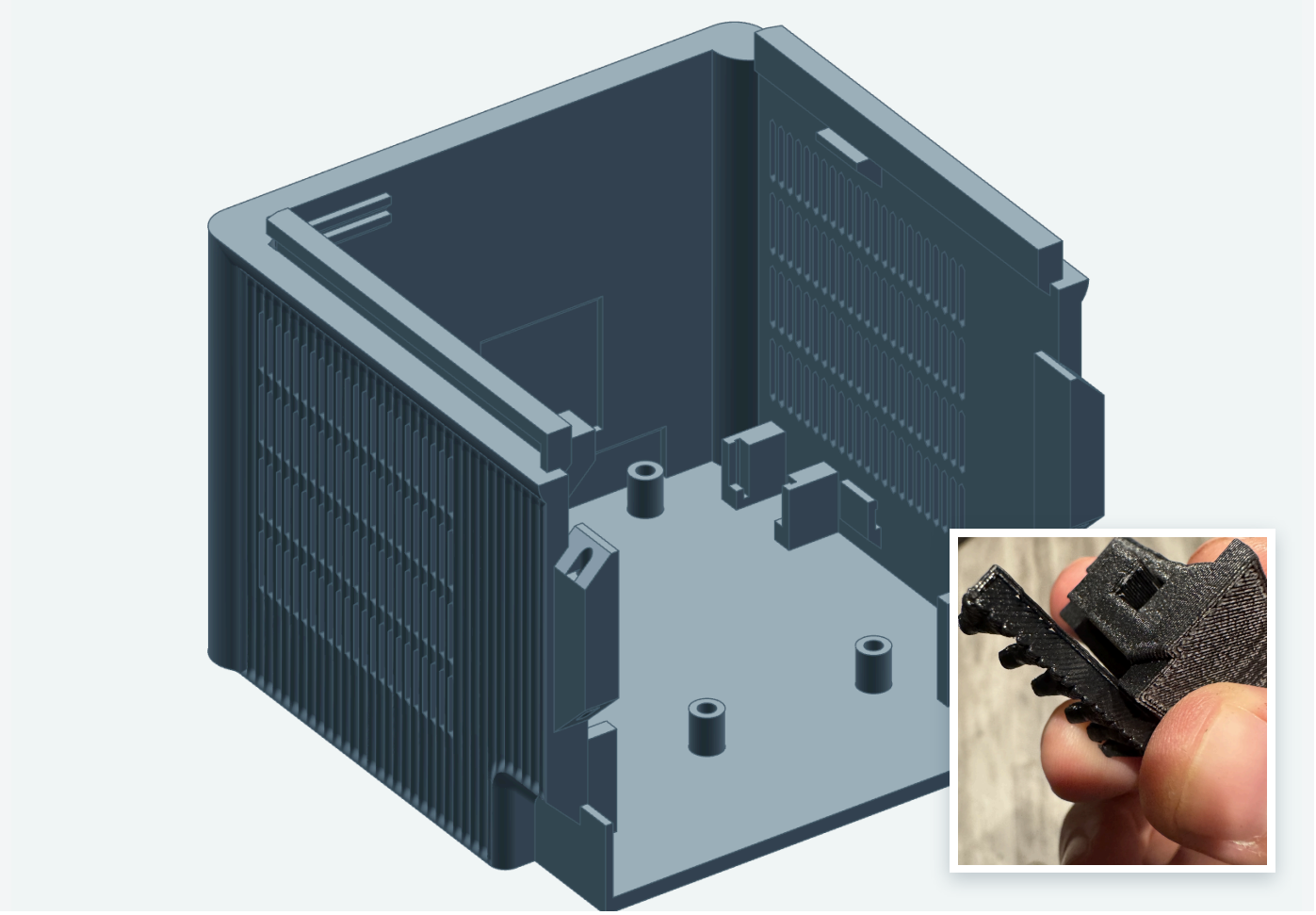
HOME SODA MACHINE

FDM print guide

How we design, slice and print the machine.



Enclosure front-bottom in its production pose. Inset shows a rail coupon and its removed support.

Geometry
Bridges, corbels and support access

Materials
Part recipes and bench results

Printer
H2C setup and first-layer checks

READING ROUTES

Start with the part

Find the feature you’re changing, then use the recipe for its material and production pose.

03 Bridges and corbels
Read the first deposited course over an opening.

04 Layer height and fit
Measure the opening that the slice produces.

05 Enclosure front-bottom
Follow a supported face from root to release.

06–07 PET-GF
Cooling, dense support and faucet geometry.

08 Clear PETG and PETG Basic
Reservoirs, lids and the part recipes.

09 TPU
Soft hollow parts, seals and travel paths.

10 ASA Aero
The float and the paused magnet insert.

11 Bambu Lab H2C
Match the plate, material and hotend to the job.

12 Find the evidence
Trace a result or record a new one.

> **At the bench**
> Follow the named part’s current recipe. Open its prepared archive to check the settings and toolpaths that apply to the feature.

During design

Read the section in print orientation. Preserve the bearing faces, clearances and routes for assembly and support removal.

During diagnosis

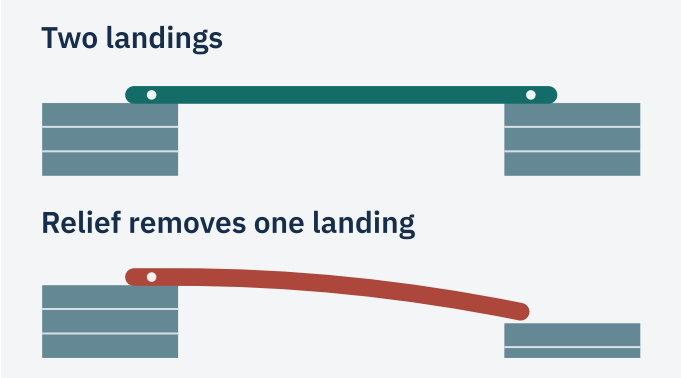
Start with the slice and the operator’s observation for that job. Use the linked trials when you need to see why a setting was chosen.

Repo guide and material index · Print workflow · Printer runbook

FUNDAMENTALS

Bridges need two landings

Trace the ends of the first spanning roads onto material that the printer has already laid down.



CROSS-SECTION SCHEMATIC

Colored lines show the first spanning course. Blue-gray courses are already on the part.



BENCH PHOTOGRAPH

First Wago lever relief. Roads drooped where the relief removed the landing.

Read the mouth perimeter

Find the first line across the opening in the native slice. Read its ends and the interior roads against the preceding courses. The Bridge label identifies a process. The geometry underneath each end tells you whether it has a landing.

Favor a short span between continuous sidewalls. Inspect where those walls meet reliefs, chamfers and corners.

Carry the slope one course at a time

A corbel advances over the course beneath it. Read the overlap along the entire feature, including at the seam and returns. The front-bottom handholds use this construction to keep their grip roofs free of support.

Keep reinforcing stock behind the seat wherever the adjoining parts and service envelope allow it.

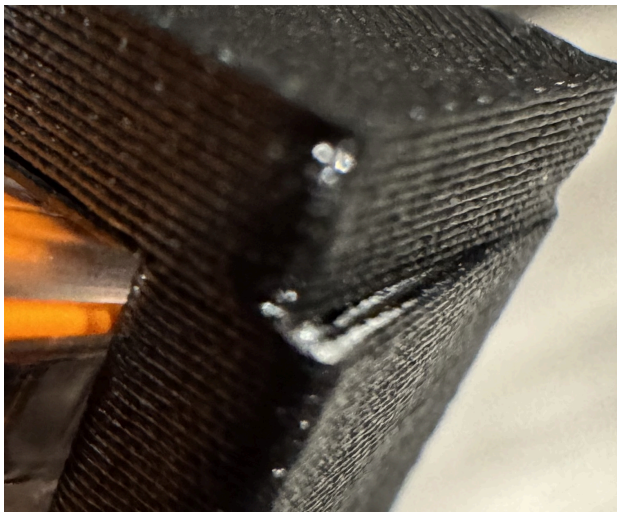
> **Rear receiver example.** In the ceiling-down back-top, the Pisco entry relief ends below a full-thickness bridge landing. The catch spans along the pocket between its end shoulders. The pocket stays clear of support where a screwdriver cannot reach.

Wago roof failure · Rear receiver access · Rear catch bridge paths

FUNDAMENTALS

Layer height is a process variable

Hold the production layer schedule when comparing fits. A small CAD change may land on the same printed course.



BENCH PHOTOGRAPH

An earlier tight PET-GF coupon split at the edge during forced insertion.

Read the actual floor and roof

Measure from the top of the last solid floor course to the underside of the first roof course. Compare the parts at the same production pose and depth.

Changing layer height also changes overlap at an overhang, the gap above support and the time before the next hot course. Use local fine layers only where the part recipe has qualified them.

The fit plate retained the 8.7 mm production insertion depth. The dots identify reductions per side from the same baseline.

Wago coupons on standard layers

Pocket and mark	CAD Z (mm)	Slice Y (mm)	Slice Z (mm)
Upper, three dots	30.0	15.8	30.0
Upper, four dots	29.9	15.6	30.0
Lower, three dots	30.2	8.4	30.24
Lower, four dots	30.1	8.2	30.0

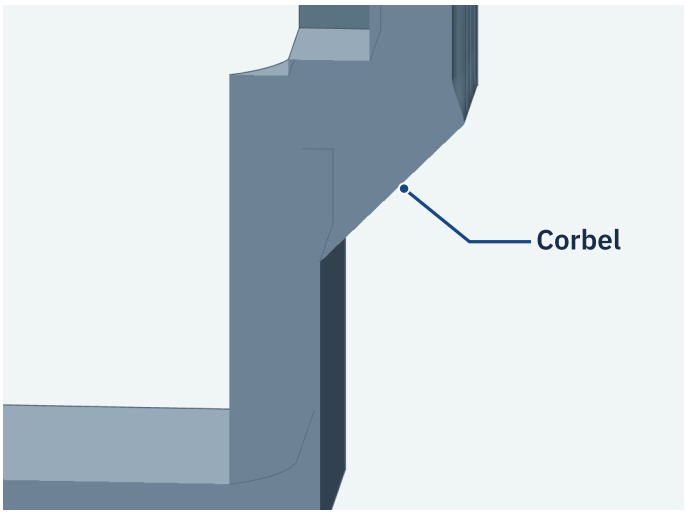
> At **0.2 mm first** and **0.24 mm ordinary** layers, both upper pockets have the same sliced Z opening. The lower pockets differ by one ordinary course. Report the shared opening and keep the test on its original schedule.

Native Wago measurements · Crack and caliper photos · Faucet fine-layer trial · Qualified enclosure layer bands

PART STUDY

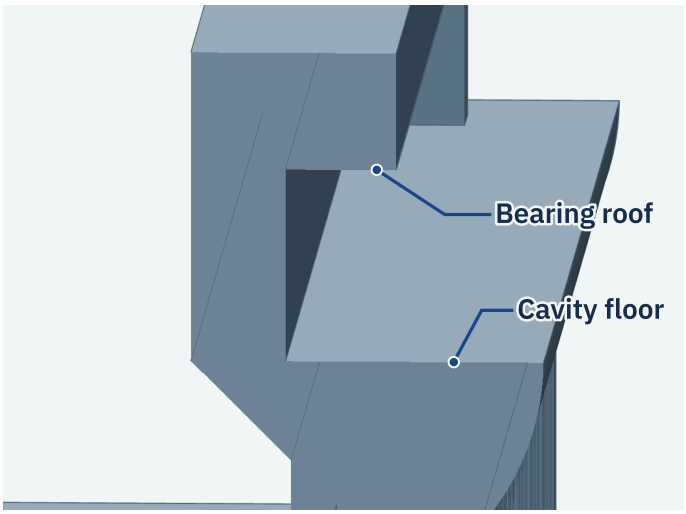
Enclosure front-bottom

Integral handhold corbels carry the gripping roofs. The two flat sliding-rail roofs get the only supports.



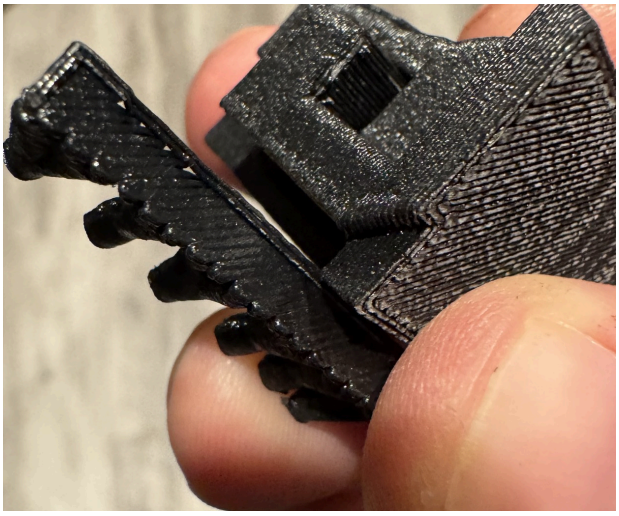
CAD SECTION · HANDHOLD

The planar corbel carries the gripping roof course by course. It continues into the full end walls.



CAD SECTION · EAST RAIL

Section across the sliding lane. The top half slides in along it and catches under the flat roof.



BENCH PHOTOGRAPH · RAIL COUPON

The support released as one piece. The mate slid into the rail.

Follow the support out

The reviewed full-part slice has **2 contacts** and **2 trees rooted on the model**. Each tree starts on a cavity floor. Lever the support free with a tool through the open sliding lane and withdraw it before assembly.

> Preserve the corbel, the full rail bearing and the open exit. Use the full-length assembly to judge clearance after a coupon fit.

Front-bottom in the machine · Full slice contact review · Full-part removal observation · Rail coupon result

Cooling follows the toolpath

Use the shared Fiberon PET-GF15 project and the named part's recipe. The part determines where it needs extra cooling.

Shared PET-GF project

Left nozzle	0.4 mm
First / ordinary layers	0.2 / 0.24 mm
Nozzle, first / ordinary	265 / 280 °C
Textured PEI bed	80 °C
Walls / gyroid	2 / 15%
Infill / wall overlap	15%
Ordinary / auxiliary fan	0% / 0%
Overhang trigger / fan	50% / 70%
Initial fan-off courses	3



BENCH PHOTOGRAPH · COVER FRACTURE

The photographed cover peeled near the rim. Derek's subsequent fan-off cover passed display loading and installation.

Cool the span and its footing

Ordinary and auxiliary cooling stay off. The project adds part fan for bridges and qualifying overhangs. The output stage retains that fan on fresh courses over the cooled footprint and caps deposition there at bridge speed.

Give a small hot tip time

Use the production batch when comparing small parts. Check the time before the nozzle returns to the same area. Where a recipe qualifies a separate tower, use its placement and retained time between visits.

Full project · Cooling and recovery policy · Cover fracture report · Fan-off cover evidence

Support D and the faucet

Use the shared dense contact recipe. The part’s geometry controls the roots, access and release.



BENCH PHOTOGRAPH · INCLINED TRIAL

The stems tore and left the dense interface on the inclined face.



BENCH PHOTOGRAPH · RIM TRIAL

The dense rim contact lifted out with the support. The insert mouths stayed open.

Shared Support D settings

Top separation	0.24 mm
Top interface courses	3
Interface line spacing	0 mm
Bridge flow ratio	1.25
Bridge speed	15 mm/s

Keep the release geometry

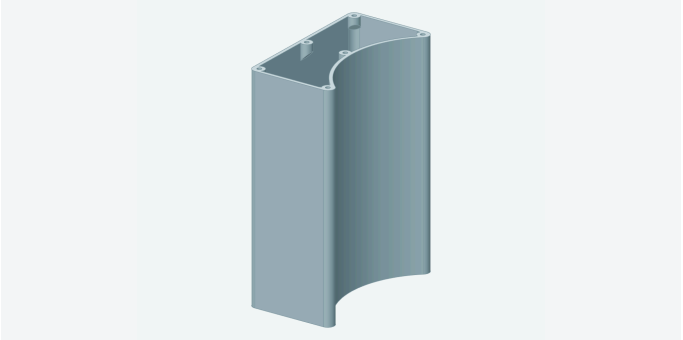
In the selected base recipes, Slim trees and segmented outer fins feed a shared dense band at the rim. The first supported outer perimeter follows the inclined rim. Use the tip’s own shoulder, dispense-face and outlet treatment.

Bridge the insert shoulders and blind pilots. Review the slice with the recesses clear for the hardware.

Support D rollout and full settings · Industrial recipe · Sculpted recipe · Inclined support result · Rim release result

Reservoirs, lids and small parts

Keep the reservoir wall and floor together in the native slice. Verify water hold on the identified article.



CAD RENDER · LEFT RESERVOIR

The floor-to-wall joint runs around the base of the body.



BENCH PHOTOGRAPH · LEAK REPORT

Derek reported a leak at the floor-fill line. The article identity and internal leak path remain open.

Clear PETG · Left reservoir

Left Standard-flow nozzle	0.8 mm
Nozzle, first / ordinary	255 / 260 °C
Glued Engineering Plate	70 °C
First / ordinary layers	0.3 / 0.24 mm
Arachne walls	5
Flow / wall overlap	1.05 / 50%
Ironing	Off

Read the floor joint

Check where solid floor fill meets the wall. The selected wall count keeps the thick curved corners in wall paths.

Reservoir lids

Check that the port cores stay open. Countersinks carry themselves, and screw seats bridge their bores. Separate TPU grips receive the tubes.

PETG Basic

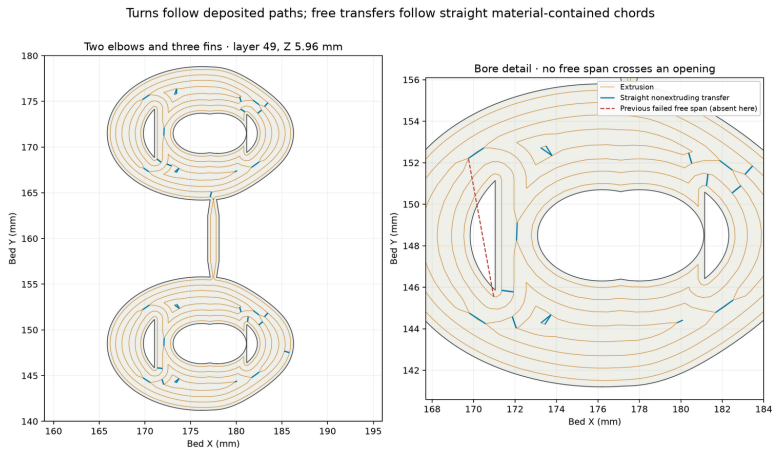
The PRV shroud prints cavity-up. The reed bridge prints on its convex face with the lead groove bridged.

> The identified left body held water at the time of Derek’s check.

Reservoir and lid recipes · Full body settings · Left water-hold result · Leak record · Ironing comparison · PETG Basic part recipes

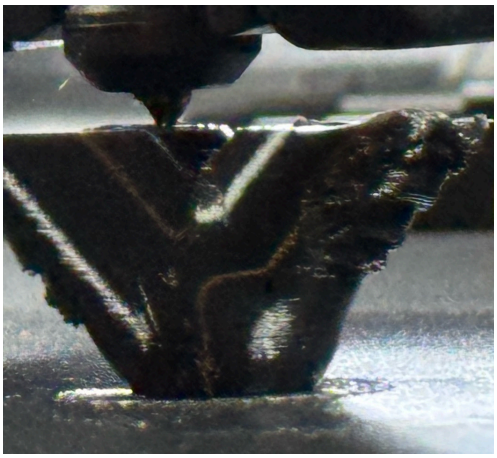
Soft walls, seals and travel

Read where a strand can attach when the nozzle leaves a wall. Short moves within deposited material keep chords out of openings.



EMITTED PATHS · SIRAYA ELBOW

The toolpath review traces the turns and links through deposited sections.



BENCH PHOTOGRAPH · EARLIER ELBOW

Distorted roads and loose loops at the underside edge.

Bambu 85A · Touch-Flo sleeve

Right nozzle	0.6 mm
Nozzle / glued Engineering Plate	225 / 35 °C
First / ordinary layers	0.3 / 0.18 mm
Maximum volumetric speed	2.2 mm³/s
Wall and cap	One spiral wall, processed planar cap

Print the processed archive

The Siraya elbow and Bambu sleeve use retained path processing. On the sleeve, the planar cap is routed before the spiral wall begins. Read its solid thickness from the emitted cap courses.

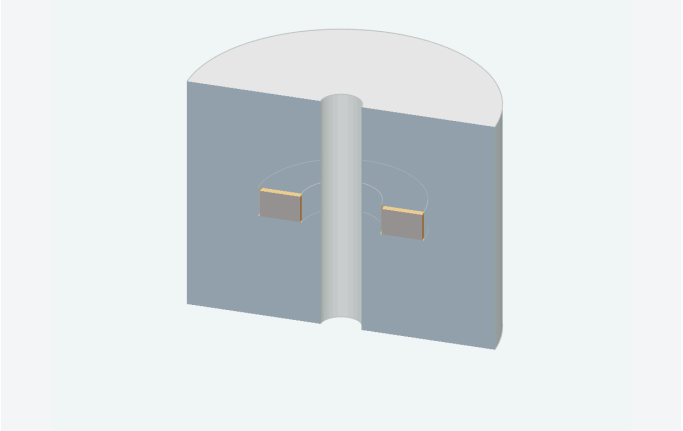
Use the stock, orientation and plate named for each flat seal, gasket and grip.

Derek reported no visible strings on the Siraya elbow and a seal that seemed as good as before. The Bambu sleeve result covers general print quality.

Elbow result · Sleeve and processed archive · White gasket recipe · Seal and grip recipes

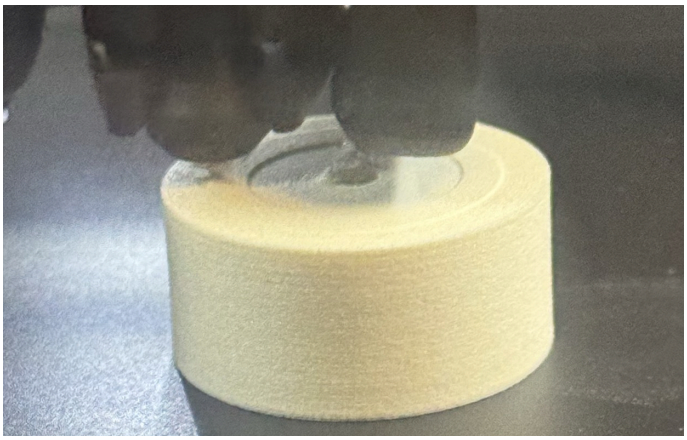
Print around a magnet

Keep the float’s guide bore open and seat the ring below the pocket rims at the programmed pause.



CAD SECTION · FLOAT

The collar beside the bore and the outer band give the covering roads their landings.



BENCH PHOTOGRAPH · PAUSED-INSERT TRIAL

Derek seated the ring and observed the first covering courses.

Selected float recipe

Nozzle	right fixed hardened standard-flow 0.4 mm
Nozzle / bed / chamber	270 / 90 / 60 °C
Flow / layers	0.52 / 0.2 mm
Nested wall loops	300
Sparse infill / skins	Off

Review the pause and roof

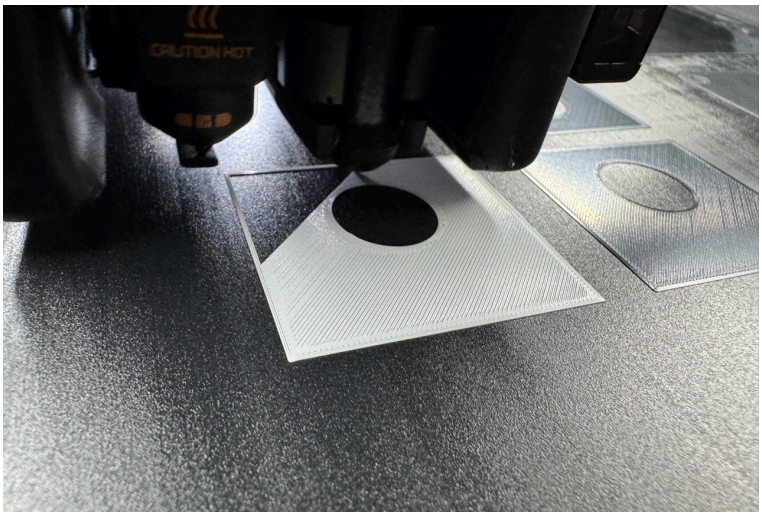
Use the recipe’s dried filament and glued Engineering Plate. Check the open pocket and first covering roads in the slice. Keep the part attached while inserting, clear loose strings and resume after the ring is seated.

Keep the heat and flow settings together because they control foaming and density. The pictured trial establishes initial overprinting. Qualify lift and reed response on the finished float.

Float recipe and dry-feed procedure · Pause and covering-path review · Overprint observation · Installed reed calibration

From plate to first layer

Mark1 and Mark2 have their own first-layer compensation. Read what the prepared archive actually commands.



BENCH PHOTOGRAPH · MARK1

The white PETG portion of the identification set on the Engineering Plate.

> **Trace the effective trim**
>
> The standard-layer Wago job emitted `G29.1 Z0.16` for Mark1 on Textured PEI. Read the prepared job and the plate-specific runbook entry for every setup.

01 **Choose the article and machine.** Match the prepared archive, current target state, hotend side and diameter, loaded spool and plate. Mark1 uses the connection key `H2C`. PET-GF may report as PET-CF. Derek's instruction to start on the named printer establishes bed clearance.

02 **Review the footprint and startup.** Check model, brim and support bead extents. Center small plates on seasoned area. Verify the effective trim in the emitted commands.

03 **Send once and read back.** Check the target and material tile in Connect or Studio. Use the job's specified send options. Follow the shared-circuit spacing before a second machine starts or resumes. Confirm the accepted task ID and archive. Watch the first model layer for adhesion and consistent roads.

04 **Keep the job with its result.** Follow the runbook's spool and extrusion recovery procedures when needed. Save the photo and operator's observation with the job.

Printer runbook · Trim table · Shared circuit · Protected printer controls · Identification-set record

REPO MEMORY

Find the evidence

Start at the repo guide. Follow a link when you need the full setting, native slice or trial behind a decision.

Geometry and recipe

Use the named part's CAD source and production STL to identify the feature. Read its saved project and preparer for the selected pose, filament, support and plate settings.

Repo guide and material index · Enclosure preparation and part recipes · Printed-part instructions

Emitted paths

Open the ready or processed `.gcode.3mf`. Read the attached review for the actual opening, support contact, bridge landing, tool or trim in question.

Rail contact example · Wago opening example

Physical result

Match the photo and operator's observation to the printer task and article. Keep the scope of the result when moving to a different pose, material or feature.

Rail fit and release example · Fitted ledger · Photo and render provenance

Add what we learned

Save the observation, original photos, source and print identity together. Link the useful conclusion from the part or material entry. Keep completed trials available at their archived Git revisions.

Repo instructions · Publish loop

> When a print contradicts a recipe, link the corrected instruction to the identified article and job that prompted it.

Geometry

Integral handhold corbels
Rail trees and cavity roots
Pisco landing and access

Material

PET-GF foundation cooling
Dense supports and bridges
Clear PETG ironing

Printer

Cancelled fine-layer plate
Shared circuit trial
H2C maintenance log